Multisensory and BIM-Integrated Digital Twin to Improve Urban Excavation Safety

Donghai Liu[1]; Chenyang Sun[2]; Junjie Chen, Aff.M.ASCE[3]; and Lei Liu[4]

Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.
Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.

Abstract: Urban excavation is an indispensable process for many construction activities such as road paving, house building, and pipe rehabilitation. However, the everincreasing complexity of underground utilities (e.g., water mains, gas lines, and sewage pipes) in urban environments challenges the safety of urban excavation, posing tremendous risks of potential collision and damage accidents. By obtaining real-time excavation information and high-fidelity simulation to evaluate safety risks, digital twin (DT) has an unexplored potential to improve urban excavation safety (UES). This research aims to investigate how a DT for urban excavation can be developed and used to improve UES. First, a multisensory solution is proposed to equip the physical excavators with the capability to precisely estimate their three-dimensional (3D) poses based on the kinematic model and social spider algorithm (SSA). Second, a building information model (BIM) of buried utilities and a 3D model of the excavators are integrated to form a dynamic virtual model that mirrors the actual excavation process. Third, based on the physical-virtual coupling DT, a real-time safety control method is proposed to proactively monitor urban excavation, dynamically assess collision risk, and timely warn against unsafe behaviors. A system prototype was developed and applied in a case study in Shandong, China. Results show that the system can precisely twin the pose of the excavator, increasing the estimation accuracy of the translation by at least 4.0 cm. The system can display the dynamic spatial position of the excavator and the buried pipes in 3D and automatically guide the excavator to operate safely in real-time, thereby avoiding potential collision accidents. **DOI: 10.1061/JCCEE5.CPENG-5354.** *© 2023 American Society of Civil Engineers.*

Author keywords: Urban excavation safety (UES); Building information modeling (BIM); Smart construction; Buried pipes; Digital twin (DT); Cyber-physical system.

Introduction

With rapid urbanization, various types of construction take place in different corners of our cities, ranging from tunneling, house development, and road renovation, to pipeline rehabilitation. Many of these construction activities involve an excavation process, which removes earth materials on the surface either in order to install/repair utilities underground, or to replace with manmade structures for foundation engineering. Excavation in urban areas has always been a challenge for the underground surface of our cities has already been filled with an extremely complex network system consisting of water mains, power lines, natural gas pipes, sewage, drainage, *inter alia*. Urban excavation among such a complex but invisible network is like "a man walking on eggshells," which, with any carelessness, can cause a collision with the underground utilities, potentially leading to destructive consequences. Over the years, there have been many major accidents in which excavators hit and destroyed underground pipes, resulting in huge loss of life and property (Li et al. 2015; Talmaki et al. 2013; Talmaki and Kamat 2012). According to the report of the Underground Pipeline Committee of China Urban Planning Association (UPCC), from 2019 to 2021, 3,450 pipeline accidents occurred in total, of which 26.1% (902 incidents) were due to excavation damage and incorrect operation, and caused 242 fatalities, 648 injuries, and hundreds of millions in property damage.

The challenges of UES reside in several aspects. First, the traditional excavation operation depends on the experience of the operator, which will lead to incorrect operation because the position of the bucket tip and the body cannot be accurately grasped. Second, due to the invisibility and uncertainty of the underground pipe, the excavator operator cannot accurately judge the relative position between the bucket tip and the underground pipe in real-time (Su et al. 2013; Tanoli et al. 2019). Third, the lack of a convenient and friendly way to accurately and timely transmit the 3D pose of the excavator and the spatial position of the underground pipe to the excavator operator.

In light of its challenging nature, research efforts have been made to improve UES. In terms of excavator 3D pose estimation, the current research estimates the 3D pose of the excavator from visual and nonvisual perspectives, respectively. Although the method of visual estimation has the advantages of low cost and convenient data acquisition, it is difficult to apply it to improve UES due to sight occlusion, camera distance limitation, and other problems (Feng et al. 2018; Liang et al. 2019; Lundeen et al. 2016; Zhao et al. 2021); The method of nonvisual estimation integrates all

[1]Professor, State Key Laboratory of Hydraulic Engineering Simulation and Safety, Tianjin Univ., 135 Yaguan Rd., Tianjin 300350, China (corresponding author). Email: liudh@tju.edu.cn

[2]State Key Laboratory of Hydraulic Engineering Simulation and Safety, Tianjin Univ., 135 Yaguan Rd., Tianjin 300350, China. Email: sun_chenyang122@tju.edu.cn

[3]Research Assistant Professor, Dept. of Real Estate and Construction, Univ. of Hong Kong, Hong Kong 999077, China. ORCID: https://orcid.org/0000-0003-4509-2271. Email: chenjj10@hku.hk

[4]Chinese Academy of Surveying and Mapping, 28, Lotus Pond Park West Rd., Beijing 100830, China. Email: rokeyliu@126.com

Note. This manuscript was submitted on January 30, 2023; approved on May 5, 2023; published online on July 4, 2023. Discussion period open until December 4, 2023; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Computing in Civil Engineering***, © ASCE, ISSN 0887-3801.

kinds of sensor information to realize 3D pose estimation of an excavator. However, the accuracy can only reach 5–10 cm (Sun et al. 2020; Talmaki and Kamat 2012; Tang et al. 2022), and there is still a possibility for further improvement. Sometimes, if the accuracy of the 3D pose estimation of an excavator is increased by 1 cm, an accident may be avoided. In terms of visual guidance systems, the current research aims to improve the UES by establishing 3D models of underground pipes and excavators and then developing intelligent guidance systems. However, the current system has the problems of poor simulation effect and insufficient accuracy of bucket tip calculation (Sharafat et al. 2021; Tanoli et al. 2018, 2019). Therefore, it is urgent to develop a safety warning technology in urban excavation to achieve high-precision 3D pose estimation of excavators. The spatial position information of underground pipes is vividly transmitted to the excavator operator in real-time to help him judge the relative position of the excavator and the underground pipe and further realize safety warnings so as to avoid pipe collision and damage accidents.

Digital twin (DT) technologies have the potential to improve collaboration between humans and excavators in urban excavation. A DT is a virtual replica of a physical entity (e.g., a product, process, system) that can exert influence on the physical counterpart by predictive analytics and simulation based on real-time collected data. This bidirectional communication mechanism and dynamic simulation capability of DT can benefit UES from various aspects. First, the DT can collect and aggregate real-time pose information of physical excavators with the capability to precisely estimate its 3D pose. Second, the DT can integrate a BIM of buried utilities, urban terrains, and a 3D model of the excavators to form a dynamic virtual model that mirrors the actual excavation process. Third, an intuitive and interactive human-robot interface enabled by the DT will proactively monitor urban excavation, dynamically assess collision risk, and timely warn of unsafe behaviors.

The objective of this research is to improve the accuracy of excavator 3D pose estimation and establish a visual, dynamic, and high-precision intelligent guidance system to ensure the safety of urban excavation. Based on the presented demand, this research proposes a digital twin-based intelligent guidance and safety warning technology for urban excavation and develops the corresponding software and hardware systems. The main contributions of this article are summarized as follows:

1. Using the Global Navigation Satellite System (GNSS) and inertial measurement unit (IMU), a multisensory solution is proposed to equip the physical excavators with the capability to precisely estimate their 3D poses based on the kinematic model and SSA.
2. Based on the Internet of Things and BIM technology, a BIM of buried utilities, urban terrains, and a 3D model of the excavators are integrated to form a dynamic virtual model that mirrors the actual excavation process.
3. Based on the physical-virtual coupling DT, the system can provide the excavator operator with the 3D pose information of the excavator and the relative distance between the excavator and the underground pipe. It further evaluates the risk of excavator operations in real-time and ensures the safety of urban excavation through early warning.

The remainder of this article is organized as follows. "Related Works" reviews the previous work related to this article. In "Methodologies," the proposed method is introduced and elaborated in detail. "System Prototype Development" develops the system. "Case Study" verifies the proposed method and developed system. "Discussion" elaborates on the research results of this article. A conclusion is drawn in "Conclusion and Prospect."

Related Works

Excavator Pose Estimation Method

In the current practical research, there are mainly two methods based on nonvisual sensors and based on a vision for excavator pose estimation.

Vision-Based Pose Estimation Method

The input information of the vision-based excavator pose estimation method often comes from the camera on the construction site. Compared with precision instruments such as laser scanners, the camera is widely used in construction sites and has the advantages of low cost and convenient data acquisition (Liang et al. 2019). The vision-based pose estimation method can directly analyze the position and orientation information from the input data, identify the on-site construction equipment (Golparvar-Fard et al. 2013; Soltani et al. 2016; Zou and Kim 2007), and then estimate the pose of its six degrees of freedom (Feng et al. 2018; Liang et al. 2019; Luo et al. 2020; Mahmood et al. 2022). According to whether the marker needs to be preinstalled on the device, it can be divided into unmarked pose estimation methods and marked pose estimation methods.

The marker-based pose estimation method uses an optical camera to monitor the fiducial markers preinstalled on the excavator manipulator and obtains the manipulator's pose through computer vision technology. Lundeen et al. (2016) developed a marker-based excavator pose estimation system, and the bucket tip depth error can be controlled within 2.5cm, but due to the complexity of the electromechanical system, the implementation on the construction site is very difficult. Feng et al. (2018) proposed a camera marker network structure in which two cameras and markers can achieve centimeter-level measurement accuracy within a range of 15 m. Zhao et al. (2021) proposed a method for estimating the pose of an excavator through a monocular camera and markers. The depth error of this method is less than 2.2cm, but it has a relatively large orientation error. Although the marker-based pose estimation method has achieved relatively good research results in terms of accuracy, this method has obvious shortcomings: on the one hand, the marker is prone to damage. On the other hand, it is difficult to keep the cameras and markers free from occlusion in a dynamic construction environment (Lundeen et al. 2016).

The unmarked pose estimation method omits the step of installing markers in advance and can directly extract pose information from images, which is convenient for application on construction sites. Soltani et al. (2017) estimated the 2D pose of the excavator by training on synthetic images of the excavator components. Traditional image processing and computer vision methods rely on the quality of the dataset, and their pose estimation bias may increase due to the constantly changing background of the construction site (Luo et al. 2020). In recent years, the combination of computer vision and convolutional neural networks (CNN) has achieved good results in estimating the 3D pose of excavators, and the decisive factor for the dramatic improvement in performance is a large, accurately labeled, and diverse dataset (Liang et al. 2019; Luo et al. 2020; Xu and Yoon 2019). However, in a practical setting, the samples that can be used to make datasets are limited, and it takes a lot of time and effort to prepare datasets with high-quality annotations (Mostafa and Hegazy 2021). Therefore, Mahmood et al. (2022) studied training a CNN on a synthetic dataset to estimate the 3D pose of an excavator, and the 3D position error of the proposed method was 51.42 cm. Using the game engine Unity, Assadzadeh et al. (2022)

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

developed a method for estimating the 3D pose of an excavator based on domain randomization (DR) to generate a labeled dataset. The model trained with synthetic data was similar to the model trained with real excavator images. Compared with the marker-based pose estimation method, the unmarked pose estimation method avoids the problem that the marker is occluded and damaged, but its position accuracy is lower, which can only reach about 50 cm. From the perspective of preventing the collision between the excavator and the underground pipe, the accuracy is difficult to meet the requirements of on-site construction (Mahmood et al. 2022). In addition, due to the contact between the bucket tip and the soil during the construction process, the shape and outline of the bucket tip in the video frame are changed, so it is difficult to accurately determine the position of the bucket tip (Assadzadeh et al. 2022; Soltani et al. 2017).

Although the vision-based excavator pose estimation method may be an efficient, low-cost, and user-friendly method, it has the following obvious drawbacks:

1. The marker-based pose estimation method has the possibility that the marker will be occluded or damaged (Feng et al. 2018; Lundeen et al. 2016; Zhao et al. 2021).
2. The 3D position accuracy of the unmarked pose estimation method cannot meet the requirements of anticollision in urban road excavation operations (Mahmood et al. 2022; Soltani et al. 2017).
3. The influence of environmental conditions such as camera distance and lighting will lead to a decrease in the accuracy of pose estimation (Feng et al. 2018; Liang et al. 2019).
4. CNN processing speed of video frames is difficult to achieve in real-time (Luo et al. 2020; Soltani et al. 2017).

In summary, the vision-based excavator pose estimation method is difficult to solve the collision detection problem of urban road excavation operations.

Nonvision-Based Pose Estimation Method

The nonvisual-based pose estimation method obtains the pose information of its mechanical components by installing monitoring equipment on the mechanical surface of the excavator. Previous studies used precision devices to obtain accurate excavator pose information. For example, Vahdatikhaki et al. (2015) combined with a genetic algorithm and proposed an excavator pose estimation method based on an ultra-wideband (UWB) real-time positioning system, improving the excavator's positioning accuracy. In comparison, IMU sensors are widely used in the industry because of their easy installation and low price. Sun et al. (2020) provided guidance for sensor selection by comparing draw-wire sensors and IMU sensors in excavator pose estimation. Talmaki and Kamat (2012) proposed a hybrid virtual excavator real-time collision detection and safety early warning framework. In this framework, they control the IMU sensor data as the input for the rotation of the excavator's mechanical components. Although the IMU sensor is not actually used in field tests, this method confirms the potential of the excavator attitude estimation based on the IMU sensor. Tang et al. (2022) proposed a method for estimating the full-body pose of an excavator based on IMU. The method uses quaternions to calculate the pose information of each link, and the 3D position accuracy can reach 5 cm.

The presented research proves the advantages of IMU sensors in excavator pose estimation. In terms of operability, IMU sensors are easy to install, low in cost, and less affected by the environment, such as object occlusion and lighting conditions. In terms of data performance, IMU data are reliable and accurate. Therefore, the excavator pose estimation method based on the IMU sensor has broad application prospects in urban road excavation operations and provides a potential solution to the problem of preventing the collision between the excavator and the underground pipe. However, the current IMU-based pose estimation is limited by the installation and calibration operation, despite its accuracy having reached 5 cm, so there is still the possibility of further improvement.

Digital Twin for Urban Road Excavation

The advancement of data acquisition technology and virtual modeling technology has promoted the development of DT technology (Madubuike et al. 2022). The DT consists of three parts, which are divided into physical models, virtual models, and their communication platforms (Akanmu et al. 2021). The emergence of DT has brought enormous opportunities to multiple industries, with its first applications in aerospace, manufacturing, and medical (Croatti et al. 2020; Liu et al. 2021; Shao and Helu 2020). In recent years, DT has shown great potential in the design, construction, operation, and maintenance of buildings, and it has also achieved many applications (Hosamo et al. 2022). Based on DT, Kaewunruen and Lian (2019) developed a management system for the sustainability-based lifecycle of railway turnouts, which can reduce construction costs and improve operational efficiency. Jiang et al. (2022) propose a general framework based on DT for the human–machine interaction construction activities. This framework is used to analyze the influence of unsafe hoisting behaviors on the stability of the tower crane. By integrating DT, deep learning (DL), and mixed reality (MR) technologies, Wu et al. (2022) develop a real-time visual alarm system that enables construction workers to proactively judge their safety status and avoid accidents.

Despite DT has been widely used in engineering construction, the research on urban road excavation based on DT is rarely involved. The current related research mainly includes the development of an excavator intelligent guidance system and the parametric modeling and visualization method of underground pipe networks. Behzadan et al. (2015) proposed to use augmented reality (AR) technology to superimpose the underground pipe network on the video frame to continuously help the excavator operator to determine the spatial position. Talmaki and Li et al. (Li et al. 2015; Su et al. 2013; Talmaki et al. 2013; Talmaki and Kamat 2012) studied the probabilistic modeling method of underground pipes and developed a simulation framework for real-time visualization of excavation operations, combined with collision detection algorithms to improve the spatial awareness of excavator operators. However, it does not establish a physical model of the excavator for on-site testing and does not use real sensor data as input. There may be certain differences in real-time and accuracy between on-site testing and simulation frameworks. Based on BIM-GIS technology, Sharafat et al. (2021) and Tanoli et al. (2018, 2019) developed an integrated framework of the underground pipe network management system, providing real-time 3D spatial information of the underground pipe network to assist the excavator operator during the construction process. Although this study combines the 3D model of the excavator and the underground pipe network to provide visual guidance and safety warning for the excavation operation, it focuses on the parametric modeling method of the underground pipe network and does not discuss the full-body pose estimation of the excavator in detail. Moon and Seo (2017) proposed a 3D modeling technology for excavators to display the pose of the excavator in a 3D virtual environment, but the error of the pose estimation reached 9cm.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Limitations of Existing Studies

Despite the advancements made by the presented studies, existing studies in the excavator pose estimation method and DT for urban road excavation have the following limitations.

1. Based on computer vision technology, the error of the unmarked excavator 3D pose estimation method is too large, and the marked method has problems such as marker occlusion and camera distance limitation. Based on the IMU sensor, although the accuracy of the excavator pose estimation method reaches 5 cm, it is difficult to accurately calibrate the IMU sensor equipment and the geometric size of the excavator, so the accuracy may be further improved.
2. For urban excavation, the current research mostly focuses on the parametric modeling method of the underground pipe network, and there are few studies on the combination of the dynamic twin model of the excavator and the 3D model of the underground pipe network for safety warning. The developed excavator's intelligent guidance and safety warning system has problems such as poor display effect and low real-time accuracy of the bucket tip.

In summary, it is necessary to combine DT to carry out intelligent guidance and safety warning for urban road excavation operations so as to realize real-time coupling of the physical model and virtual model of the excavator and high-precision 3D pose estimation and then realize real-time safety warning and remote monitoring of urban excavation operations to ensure construction safety.

Methodologies

In view of the shortcomings of previous research, this research proposes an intelligent guidance and safety warning technology for urban excavation operations based on DT, which aims to solve the problem of collision damage between excavators and underground pipes. Fig. 1 shows the workflow of the proposed technique, which consists of three parts: (1) physical model, (2) virtual models, and (3) real-time guidance and security warning.

Physical Model

Multisensory Hardware Setup

The hardware device of the physical model mainly includes four IMU sensors, two GNSS receivers, a data controller, and a vehicle navigation screen. The installation location is shown in Fig. 2(a). IMU sensors collect the pose data and GNSS receivers collect the position data of the excavator in real-time. The data controller is a bridge between the physical model and the virtual model and realizes dynamic coupling by transmitting the data of the sensor and the GNSS receiver to the virtual model in real-time. The vehicle navigation screen displays the system information of intelligent guidance and safety warning to the excavator operator in real-time and gives warnings to dangerous areas.

Pose Data Sensing

As shown in Fig. 2(b), the pose data collected from the physical model can be divided into two types, one is the real-time data collected by the IMU sensor and the GNSS receiver, and the other is the mechanical structure size data of the excavator. The four IMU sensors are used to obtain the pose data of the body, the boom, the arm, and the bucket, respectively. Based on this data, the 3D pose of the excavator can be estimated. The two GNSS receivers obtain the 3D coordinates and heading angle of the excavator respectively, and then the 3D pose coordinates of the excavator can be converted to the World Geodetic System (WGS)-84 coordinate system

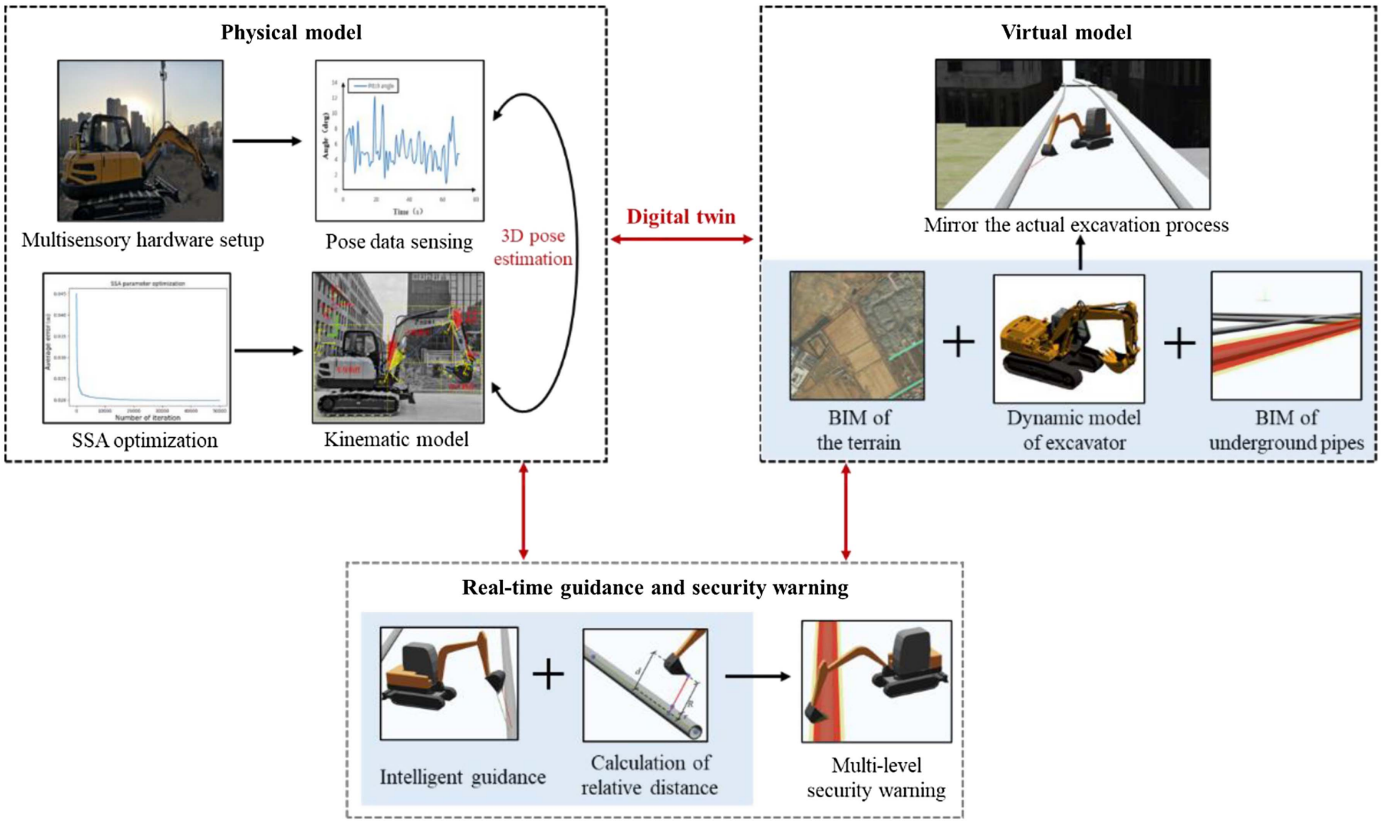


Fig. 1. Workflow of DT-based excavator intelligent guidance and safety warning method. (Images by Chenyang Sun.)

J. Comput. Civ. Eng., 2023, 37(5): 04023025

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

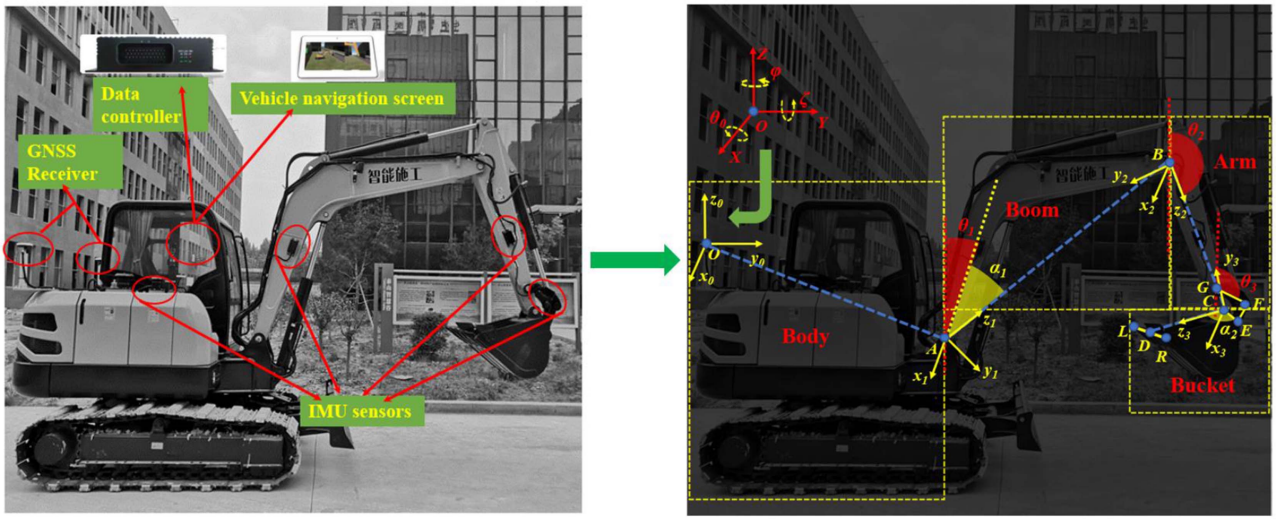


Fig. 2. Data collected by excavator intelligent guidance device and kinematic model of 3D pose estimation: (a) excavator intelligent guidance device; and (b) pose data and geometric relationships. (Images by Chenyang Sun.)

through coordinate conversion. The mechanical structure size data of the excavator are used for 3D pose estimation and the establishment of the dynamic mapping of the excavator. Taking point O as the origin of the coordinate system, the lateral direction of the body is the x-axis, and the longitudinal direction is the y-axis, conforming to the right-hand rule of the Cartesian coordinate system to establish the body coordinate system. The pose data of the excavator are shown in Table 1.

3D Pose Estimation Based on SSA

In this section, first, based on the Euler coordinate transformation method, combined with the mechanical structure characteristics and the collected pose data, a kinematic model of the excavator was established to realize the 3D pose estimation of each mechanical component, and then, based on SSA, the geometric parameters in the calculation process of the bucket tip coordinates are optimized to improve the accuracy.

Table 1. The pose data of the excavator

Name	Details
Real-time data	
θ_0	The vertical angle of the body
θ_1	The vertical angle of the boom
θ_2	The vertical angle of the arm
θ_3	The vertical angle of the bucket
ξ	Roll angle of the excavator
φ	Heading angle of the excavator
(X_0, Y_0, Z_0)	The 3D coordinates of point O in the WGS-84 coordinate system
Mechanical structure size data	
(x_1, y_1, z_1)	The 3D coordinates of point A in the body coordinate system
L_{AB}	Boom length
L_{BC}	Arm length
L_{CD}	Bucket length
L_{CE}, L_{EF}, L_{FG}, L_{GC}	Bucket moving link length
L_{DL}, L_{DR}	Left and right offset length of bucket tip
α_1, α_2	Boom and bucket offset angle

Step1: Kinematic Model of Excavator Based on Euler Transformation. Based on the pose data collected in the section "Pose Data Sensing," a kinematics model of the excavator is established using forward kinematics to calculate the three-dimensional coordinates of the excavator bucket tip control points in the real world. As shown in Fig. 2(b), the excavator is simplified into four components, namely, the body, the boom, the arm, and the bucket. The calculation sequence is O-A-B-C-D, where the starting point O is the coordinate of the main GNSS receiver, and the ending point D is a bucket tip control point. The detailed calculation process is as follows:

1. Establish the space rectangular coordinate system of each component.

   The establishment of the coordinate system of all components conforms to the right-hand rule of the Cartesian coordinate system in order to facilitate the subsequent coordinate transformation. As shown in Fig. 2(b), the body coordinate system takes point O as the origin, the lateral direction of the body is the x-axis, and the longitudinal direction is the y-axis. The boom coordinate system takes point A as the origin, the lateral direction of the body is the x-axis, and L_{AB} is the z-axis. The arm coordinate system takes point B as the origin, the lateral direction of the body is the x-axis, and L_{BC} is the z-axis. The bucket coordinate system takes point C as the origin, the lateral direction of the body is the x-axis, and the L_{CD} is the z-axis.
2. Determine the 3D coordinates of the target control points of each component in its own coordinate system.

   All components are rigid bodies, and their coordinate systems are independent of each other. Therefore, the 3D coordinates of each component target control point in its own coordinate system will not change with the movement or rotation of the component, so this method can realize the 3D coordinate transformation of any point on the component. As shown in Fig. 2(b), this research selects point A as the target control point of the body, and the 3D coordinate of point A in the body coordinate system is (x_1, y_1, z_1). Point B is selected as the target control point of the boom, and the 3D coordinate of point B in

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

the boom coordinate system is $(0, 0, L_{AB})$. Point C is selected as the target control point of the arm, and the 3D coordinate of point C in the boom coordinate system is $(0, 0, L_{BC})$. For the convenience of explanation, this research takes point D as the target control point of the bucket, and the 3D coordinate of point D in the bucket coordinate system is $(0, 0, L_{CD})$.

3. Calculate the 3D coordinates of the target control points of each component in the WGS-84 coordinate system by Euler coordinate rotation.

As the simplest way to express attitude, the Euler angle can express rigid body attitude with only three elements. It has a wide range of applications in aerospace (Zeng and Guo 2017), animation (Perumal 2014), and human pose estimation (Huang et al. 2022; Panchal and Shah 2013; Sun and Lu 2020). Depending on the order of the rotation axis, there are as many as 12 expressions for Euler angles (Perumal 2014). This research selects the z–y–x Euler angle for calculation, first rotating the heading angle around the z-axis, then rotating the roll angle around the y-axis, and finally rotating the pitch angle around the x-axis, so as to realize the conversion from the WGS-84 coordinate system to the component coordinate system. Because the roll angle of the excavator cannot be 90° during the actual excavation operation, this rotation sequence can avoid the singularity associated (Zeng and Guo 2017).

According to the 3D coordinates of each target control point in its own coordinate system, the 3D coordinates under the WGS-84 coordinate system can be calculated by Eqs. (1) and (2)

$$\begin{bmatrix} X_{tar} \\ Y_{tar} \\ Z_{tar} \end{bmatrix} = K^{-1} \begin{bmatrix} x_{tar} \\ y_{tar} \\ z_{tar} \end{bmatrix} + \begin{bmatrix} X_0 \\ Y_0 \\ Z_0 \end{bmatrix} \tag{1}$$

$$K = \begin{pmatrix} \cos\zeta\cos\varphi & \sin\phi\sin\zeta\cos\varphi - \cos\phi\sin\varphi & \cos\phi\sin\zeta\cos\varphi + \sin\phi\sin\varphi \\ \cos\zeta\sin\varphi & \sin\phi\sin\zeta\sin\varphi + \cos\phi\cos\varphi & \cos\phi\sin\zeta\sin\varphi - \sin\phi\cos\varphi \\ -\sin\zeta & \sin\phi\cos\zeta & \cos\phi\cos\zeta \end{pmatrix} \tag{2}$$

where $(X_{tar}, Y_{tar}, Z_{tar})$ = 3D coordinate of the target control point in the WGS-84 coordinate system; and $(x_{tar}, y_{tar}, z_{tar})$ =3D coordinate of the target control point in the component coordinate system. (X_0, Y_0, Z_0) is the 3D coordinate of the component origin in the WGS-84 coordinate system, calculated from the previously connected component, where the origin coordinate of the body component is the WGS-84 coordinate of point O, which is obtained by the GNSS receiver. K is the rotation matrix based on Euler angles, where φ, ξ, ϕ are the heading angle, roll angle, and pitch angle of the component, respectively. According to the geometric characteristics of the excavator, in the process of coordinate conversion of each component, the roll angle and heading angle are kept consistent with the excavator, the heading angle is calculated by the GNSS receiver, and the roll angle is obtained by the IMU sensor on the body. The pitch angles of each component can be obtained by θ_0, θ_1, θ_2, and θ_3 and the mechanical structure size data of the excavator.

Step2: Bucket Tip Coordinate Optimization Based on SSA. The error of the 3D pose estimation of the excavator is mainly caused by the following three reasons. (1) Random errors of GNSS receivers and IMU sensors. (2) Systematic errors of GNSS receivers and IMU sensors. (3) Some target control points may be inside the component, which are difficult to measure directly, so there is a measurement error in the mechanical structure size data. Therefore, an intelligent optimization algorithm can be used to calibrate the geometric parameters in the solution of the bucket tip coordinates to improve the accuracy.

This research ignores the random error of hardware equipment and considers that the 3D coordinates measured by the laser total station are the real spatial position of the bucket tip and the point D of the excavator bucket tip, which is the most critical in urban excavation, is selected for optimization. On the one hand, the occurrence of urban excavation accidents is often caused by the collision between the bucket tip and the underground pipe. On the other hand, point D is the end point of the 3D pose estimation, and its error is the accumulation of the calculation error of the previous component's target control point. Therefore, the calculation error of point D is the largest, and the 3D coordinate estimation error of the target control point of the remainder component is smaller than it.

The bucket tip coordinate optimization model proposed in this paper is a single-objective optimization model. The optimization goal is that the point D coordinate obtained by the 3D pose estimation method is as close as possible to the coordinate measured by the laser total station. The geometric parameters in the 3D pose estimation of point D are optimized. As shown in Table 2, there are a total of 18 optimization variables, and the size variables and angle variables are adjusted within a certain range, respectively.

After the bucket tip coordinate optimization model has been established, the intelligent optimization algorithm used in this paper is SSA, which has the advantages of a simple structure, few parameters, and fast convergence (Yu and Li 2015). As a new global intelligent optimization algorithm, its idea originates from

Table 2. Optimization variables for the bucket tip

Size variables	Angle variables
x_1	α_1
y_1	α_2
z_1	φ
L_{AB}	ξ
L_{BC}	θ_0
L_{CD}	θ_1
L_{CE}	θ_2
L_{EF}	θ_3
L_{FG}	—
L_{GC}	—

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

the natural phenomenon that spiders send information to their companions by vibrating their webs after finding food, thereby attracting other spiders to come for food. At present, SSA is widely used in power dispatching optimization (Adhvaryyu and Adhvaryyu 2020; El-bages and Elsayed 2017), train transportation (Sung et al. 2019), neural network training (Gülmez and Kulluk 2019), and so on. The algorithm flow of the SSA method to optimize the presented model is as follows:

1. Fitness function

   In SSA, the optimization space is simulated as a hyperdimensional spider web, and each position of the spider web represents a feasible solution to the optimization space. Initialize the position of the spider population, and calculate the fitness of each spider by Eq. (3)

$$f(P_s) = \frac{1}{N}\sum_{i=1}^{N}\sqrt{[(x_{s,i}-x_{0,i})^2+(y_{s,i}-y_{0,i})^2+(z_{s,i}-z_{0,i})^2]} \tag{3}$$

   where $f(P_s)$ = fitness of the spider s; and $(x_{s,i}, y_{s,i}, z_{s,i})$ = bucket tip coordinate of spider s in sample i calculated by Euler transformation. $(x_{0,i}, y_{0,i}, z_{0,i})$ denotes the bucket tip coordinate measured by the laser total station of sample i, and N is the number of samples.

2. Vibration generation and propagation

   When the spider s reaches a new position, it will emit a vibration at the current position. The intensity of the vibration at the source position is related to the fitness of the spider, which is calculated by Eq. (4)

$$I(P_s, P_s, t) = \ln\left(\frac{1}{f(P_s)-c}+1\right) \tag{4}$$

   where $I(P_s, P_s, t)$ = vibration intensity at the source position of the spider s at time t, which is related to $f(P_s)$. This research considers a minimization problem, so the larger $f(P_s)$ is, the smaller the vibration intensity is. c is a very confident small constant, ensuring that all vibration intensities are greater than zero.

   After the vibration is generated, it will spread throughout the spider web and be sensed by other spiders, and the vibration intensity will attenuate as the distance increases between spiders. Through the Manhattan distance calculation method, the relative distance between spider a and spider b can be obtained from Eq. (5). Further, the vibration intensity of spider b felt by spider a at time t is calculated by Eq. (6)

$$D(P_a, P_b) = \|P_a - P_b\|_1 \tag{5}$$

$$I(P_a, P_b, t) = I(P_a, P_a, t) \times \exp\left(-\frac{D(P_a, P_b)}{\bar{\sigma} \times r_a}\right) \tag{6}$$

   where σ = standard deviation of the spider population in each dimension; and r_a = vibration attenuation coefficient, and the larger r_a means the weaker the vibration attenuation.

   Each spider s receives vibration generated by other spiders, including source location and attenuation intensity. Spider s selects the strongest vibration V_s^{best} and compares its magnitude with V_s^{tar}. If the intensity of V_s^{best} is high, then spider s will store V_s^{best} as the new V_s^{tar} and resets c_s to zero. Otherwise, retain the V_s^{tar} and increment c_s by 1.

3. Search pattern

   The algorithm uses a dimension mask to manipulate spider s to perform random walks toward V_s^{tar}. Each spider corresponds to a dimension mask M, which is a 0–1 binary vector of length D, and D is the dimension of optimization. Initially, all the values in mask M are zero, and in each iteration, spider s has a probability of $1 - p_c^{c_s}$ to change its mask, where $p_c \in (0, 1)$ is a user-defined attribute that describes the probability of changing masks. If it is decided to change the mask, each bit of the vector has a probability of p_m to be assigned with 1 and $1 - p_m$ to be 0. p_m is also a user-controlled parameter defined in (0, 1). Each bit of a mask is changed independently and does not have any correlation with the previous mask. After the dimension mask is determined, a new following position $p_s^{f_0}$ is generated based on the mask for s. The value of ith dimension of the following position $p_{s,i}^{f_0}$ is generated by Eq. (7). s performs a random walk to the position $p_s^{f_0}$. This random walk is conducted by Eq. (8)

$$p_{s,i}^{f_0} = \begin{cases} p_{s,i}^{tar} & m_{s,i} = 0 \\ p_{s,i}^{a} & m_{s,i} = 1 \end{cases} \tag{7}$$

$$p_s(t+1) = p_s + (p_s - p_s(t-1)) \times r + (p_s^{f_0} - p_s) \odot R \tag{8}$$

   where a = random integer value generated in $[1, |pop|]$; $m_{s,i}$ = ith dimension of the dimension mask M of spider s; r = random number in (0,1); $\odot$ = element-wise multiplication; and R = vector of random float-point numbers generated from zero to one uniformly.

   Spiders may walk out of the spider web every time. For those spiders that do not meet the constraints, the boundary constraints are carried out by the reflecting algorithm, which is calculated by Eq. (9)

$$p_{s,i}(t+1) = \begin{cases} (\bar{x}_i - p_{s,i}) \times r & p_{s,i}(t+1) > \bar{x}_i \\ (p_{s,i} - \underline{x}_i) \times r & p_{s,i}(t+1) < \underline{x}_i \end{cases} \tag{9}$$

   where $\bar{x}_i$ and $\underline{x}_i$ = upper and lower bounds of the search space in the ith dimension, respectively.

Virtual Model

The virtual model mirrors the actual excavation process, including the BIM of underground pipes and the dynamic mapping of excavators.

BIM of Underground Pipes

On the one hand, the BIM model of underground pipes presents the spatial position of underground pipes to the excavator operator in real-time. On the other hand, it is used to calculate the relative distance between the bucket tip and the underground pipes in real-time to ensure the safety of urban excavation. As shown in Fig. 3, the BIM model of the underground pipes is established according to the design data or pipes data collected by ground penetrating radar (GPR). Each underground pipe is uniquely determined by the initial 3D coordinates of the central axis of the pipe and the pipe diameter. At the same time, considering the uncertainty of recorded data, two layers of safety warning areas are set outside the pipes during modeling. The size of the area is related to the accuracy of underground pipes data. In this research, high-risk areas and low-risk areas are set 30 cm and 50 cm away from the outer wall of the pipe respectively. Considering the real-time issue of the subsequent calculation of the relative distance between the bucket tip and the pipes, the established model is segmented every 100 meters according to the mileages.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

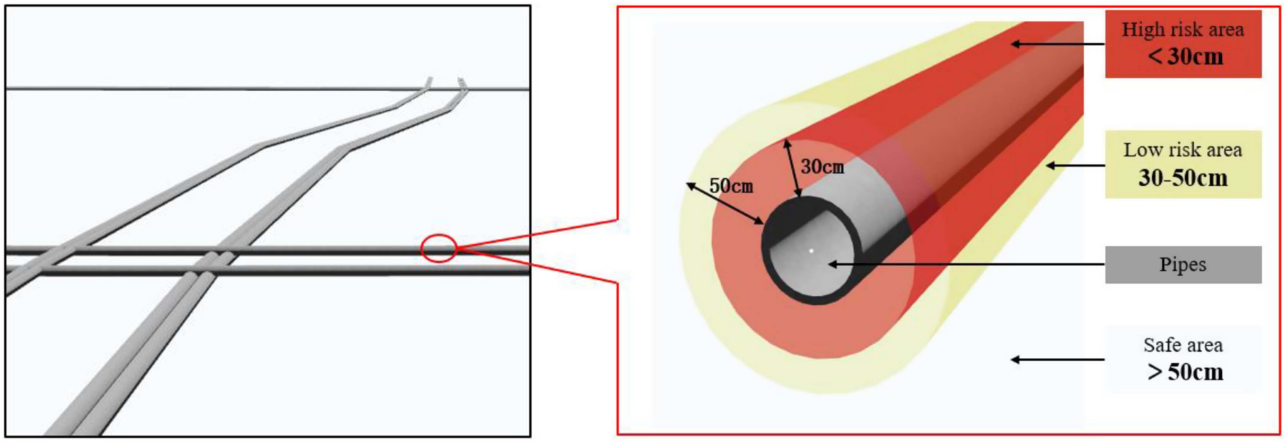


Fig. 3. BIM of underground pipes.

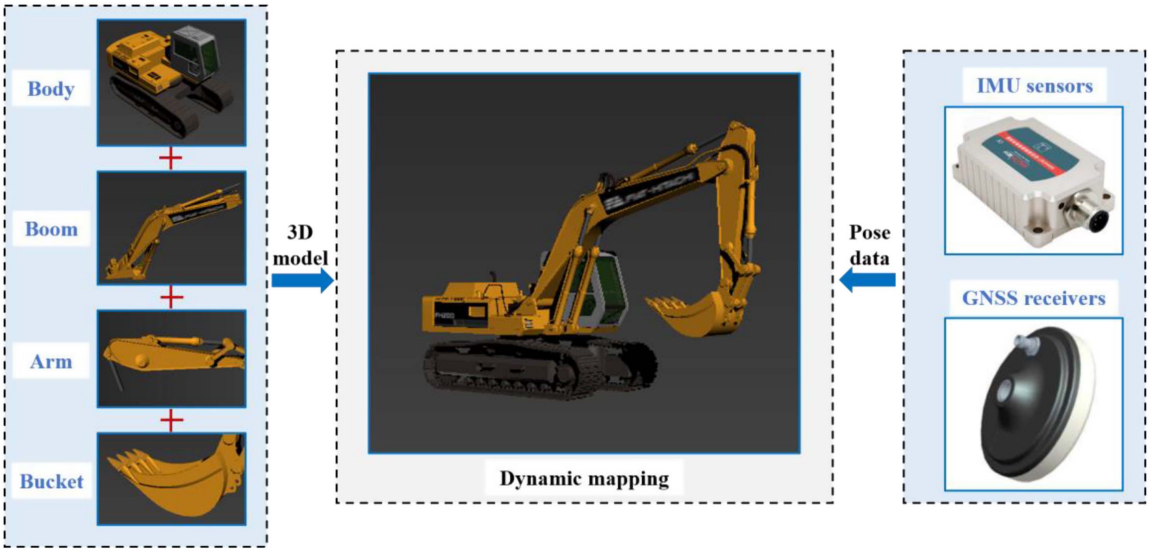


Fig. 4. Dynamic model of the excavator.

Dynamic Mapping of Excavator

The dynamic mapping of excavators can realize the real-time coupling between the virtual model and the physical model and visually presents the 3D position and pose to the excavator operator. Besides, it is used to judge the relative distance between the bucket tip and the underground pipe in real-time. As shown in Fig. 4, the 3D model of the excavator is established according to the mechanical structure size data collected in the section "Pose Data Sensing." With Unity as the 3D visualization framework, the 3D model of the excavator receives data of the IMU sensor and GNSS receivers in real-time and estimates the 3D pose of the excavator by the method described in the section "3D Pose Estimation Based on SSA." Then, the relative distance between the excavator and the underground pipe can be calculated in real-time to realize the safety warning of urban excavation and ensure the safety of excavation.

Real-Time Guidance and Safety Warning

Real-Time Calculation of the Relative Distance between Bucket Tip and Underground Pipes

Before calculating the relative distance, it is necessary to judge the mileage of the bucket tip according to its coordinate and then select the BIM model of the underground pipes in this area for calculation. The BIM model of the underground pipes is used to obtain the starting and ending coordinates of the pipe axis and the pipe radius, and the 3D coordinates of the bucket tip of the excavator are obtained in real-time through the 3D pose estimation method. The relative position between the bucket tip and the central axis of the underground pipes is calculated by Eqs. (10)–(13)

$$\frac{x-x_1}{m}=\frac{y-y_1}{n}=\frac{z-z_1}{p}=k \tag{10}$$

$$m(x-x_0)+n(y-y_0)+p(z-z_0)=0 \tag{11}$$

$$k=\frac{m(x_0-x_1)+n(y_0-y_1)+p(z_0-z_1)}{m^2+n^2+p^2} \tag{12}$$

$$d=\sqrt{(mk+x_1-x_0)^2+(nk+y_1-y_0)^2+(pk+z_1-z_0)^2} \tag{13}$$

where (x_1, y_1, z_1) = starting coordinate of the central axis of the underground pipes; and (m, n, p) = direction vector of the central axis. (x_0, y_0, z_0) is the 3D coordinate of the bucket tip of the

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

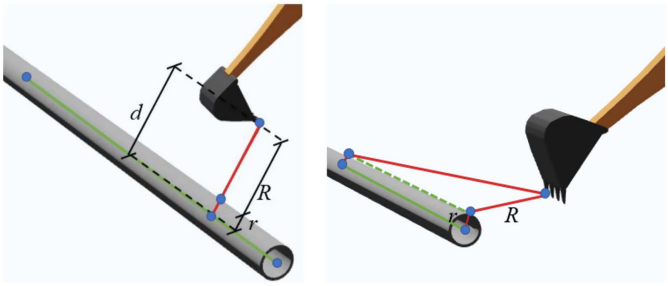


Fig. 5. Relative distance between bucket tip of excavator and underground pipes: (a) vertical foot point on the pipe; and (b) vertical foot point on the pipe extension line.

excavator. d is the vertical distance between the bucket tip and the central axis.

If the vertical foot point is located on the central axis, as shown in Fig. 5(a), the relative distance between the bucket tip and the underground pipes is calculated by Eq. (14). If the vertical foot point is on the extension line of the central axis of the underground pipe, as shown in Fig. 5(b), the relative distance is the minimum distance from the bucket tip to the end point of the outer wall of the underground pipes

$$R = d - r \tag{14}$$

where R = relative distance between the bucket tip of the excavator and the underground pipes; and r = radius of the underground pipes.

Steps of Real-Time Guidance and Security Warning

Multilevel safety warning mode is adopted for urban road excavation, and the specific steps are as follows:

1. Spatial division

   According to the BIM model of the underground pipe network in the section "BIM of Underground Pipes," the working environment is divided into three spatial areas: high-risk area, low-risk area, and safe area. Based on engineering experience, the high-risk area is the space area 0 to 30 cm away from the outer wall of the underground pipes. The low-risk area is the space area 30 to 50 cm away from the outer wall of the underground pipes. The safety zone is set to be more than 50 cm away from the outer wall of the underground pipes.
2. Spatial area determination

   According to the calculation method of the relative distance between the bucket tip and the underground pipes in the section "Real-Time Calculation of the Relative Distance between Bucket Tip and Underground Pipes," calculate the relative distance R between the bucket tip and the underground pipes, and further identify the spatial area where the bucket is located in real-time. When R is greater than 50cm, it indicates that the excavator bucket is in a safe area. When R is greater than 30cm and less than 50cm, it indicates that the excavator bucket has entered the low-risk area. When R is less than 30cm, it indicates that the excavator bucket has entered a high-risk area.
3. Multilevel security warning

   The safety warning is carried out according to the different space areas where the excavator bucket is located. When the excavator bucket is located in the safety area, the operator can work with confidence. When the excavator bucket is located in the low-risk area, the voice will remind the excavator operator to be careful with the construction, and the BIM model of the underground pipes will turn yellow. When the excavator bucket is located in the high-risk area, the BIM model of the underground pipes will turn red and the excavator operator should stop the excavation operation for detection to avoid a collision between the excavator tip and the underground pipes.

System Prototype Development

Fig. 6 is the architecture diagram of intelligent guidance and safety early warning systems for urban excavation. The system framework is divided into four parts, which are the vehicle terminal, database and application servers, remote monitoring client, and field managers.

The vehicle terminal is divided into a physical model, virtual model, and data transmission module. The physical model consists of an excavator, two GNSS receivers, four IMU sensors, a data controller, and a Pad. The data controller has a multifunction port and a built-in data transmission unit (DTU). The multifunction port includes a power port, an RS-232 serial port, a CAN serial port, and a GNSS antenna port. The power port is used to connect the excavator power supply module, and RS-232 serial port is used to connect the Pad. The CAN port is used to connect the IMU sensors, and the antenna port is used to connect the GNSS receivers. Considering the real-time performance of subsequent calculations, we use wired transmission protocols for data transmission between various hardware devices of the vehicle terminal, instead of using wireless local area networks (WLANs) or public networks. The data controller obtains tasks from servers through DTU, including work area, work time, and model information, integrates GNSS receivers data and IMU sensors data, and transmits the above data to Pad in real-time through RS-232 serial port to improve the response speed. The vehicle terminal adopts a Client/Server architecture, and the Pad can directly perform calculations and analysis after receiving data. Unity is used as the 3D visualization frame to establish the virtual model on the Pad, which can realize the 3D pose estimation, relative distance calculation, real-time safety warning, and visualization guidance. The Pad visually displays the calculation results and 3D models of the excavator and its surrounding scenes to the excavator operator in real-time to guide the excavation operation. The results are transmitted back to the data controller in real-time. The data controller transmits the above data to the server through DTU.

The server receives the returned data, which is further stored and analyzed. The remote monitoring client adopts a Browser/Server architecture and implements remote three-dimensional visualization based on Three.js. The remote monitoring client should be prepared for monitoring first, including establishing a BIM model and setting excavation tasks. Meanwhile, it realizes remote real-time monitoring and safety warning for excavation work, helping the system administrator judge the operation situation of the entire work area. After excavation, the system administrator can inquire about the warning information and excavation trajectory and playback the excavation process on the remote monitoring client to judge the cause of the warning. Personal digital assistants (PDAs) can receive warning messages and task information sent by servers to assist field managers in controlling.

Case Study

Experiment Design and Result Analysis of 3D Pose Estimation

The GNSS receivers and IMU sensors are installed and debugged on-site. Both of them collect position and pose data at 20Hz.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

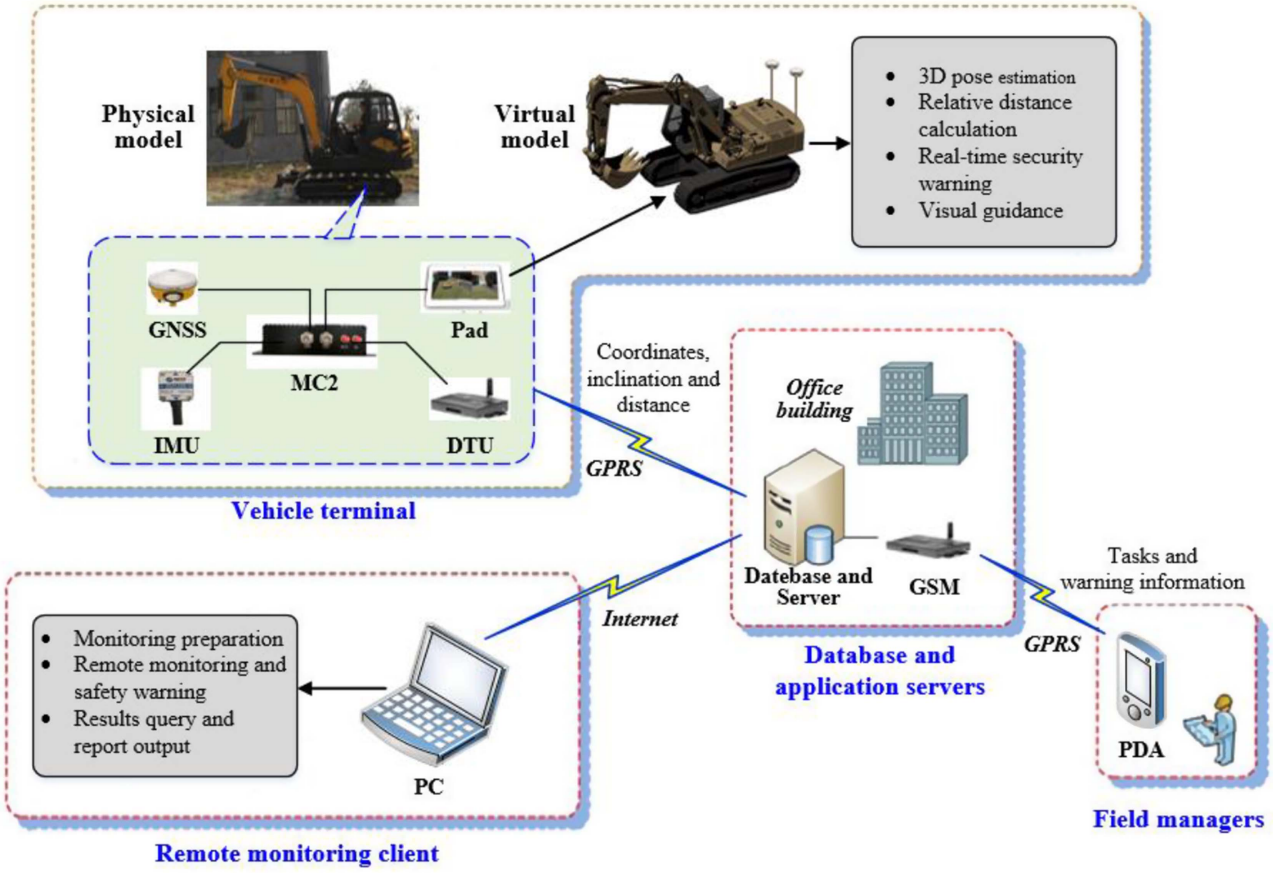


Fig. 6. Framework of intelligent guidance and safety warning system for urban Excavation.

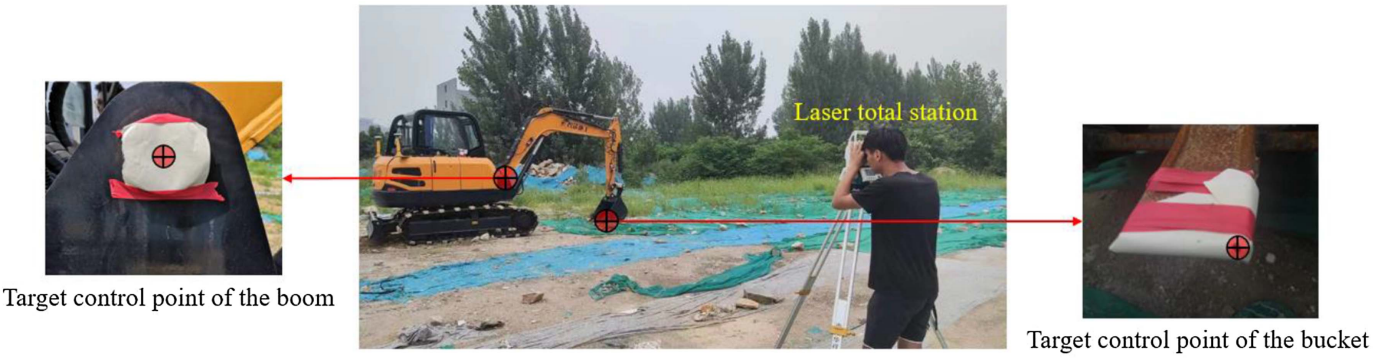


Fig. 7. Field measurement of mechanical structure size data and bucket tip coordinates.

The collected data are sent to Pad in real-time through the data controller, and the mechanical structure size data of the excavator are collected by laser total station, as shown in Fig. 7.

Geometric parameters of excavator 3D pose estimation are optimized based on SSA. The 90 groups of sample data collected are divided into training and test datasets. The training dataset contains 80 sets of data for establishing an optimization model for bucket tip coordinate. The test dataset predicts and evaluates the model, which contains 10 sets of data. In the solution of SSA, the size of the spider population is 25 and the number of iterations is 50,000. We adopt the following parameter combination: a vibration attenuation coefficient of $r_a = 1.0$ and two probability parameters of $p_c = 0.7$ and $p_m = 0.1$. The size and angle variables are changed in the range of 5 cm and 5°, respectively.

The results of the optimization of excavator mechanical structure parameters are shown in Table 3. The GNSS receiver and the IMU sensor parameters are optimized as shown in Table 4. By comparing the results of 3D pose estimation of the excavator bucket tip before and after the parameter optimization of the training dataset, as shown in Fig. 8(a), it indicates that there is a certain systematic error in the calculation results before optimization, with the mean error of 47.60 mm. And after optimization, the systematic calculation error is significantly reduced, with a mean error of 19.83 mm. The model after parameter optimization is evaluated. As shown in

Table 3. Optimized results of excavator mechanical structure parameters

Parameter	Before optimization	After optimization
x_1/mm	−360.00	−403.62
y_1/mm	1,030.00	1,016.97
z_1/mm	−510.00	−501.62
L_{AB}/mm	2,992.00	3,011.83
L_{BC}/mm	1,597.00	1,604.04
L_{CD}/mm	927.00	900.38
L_{CE}/mm	260.00	269.90
L_{EF}/mm	317.00	305.40
L_{FG}/mm	351.00	362.58
L_{GC}/mm	241.00	237.14
α_1/degrees	35.60	35.47
α_2/degrees	104.87	107.93

Table 4. Optimized results of the GNSS receiver and the IMU sensor parameters

Parameter	Correction value (degrees)
φ	0.94
ξ	0.76
θ_0	−2.58
θ_1	−0.39
θ_2	−0.06
θ_3	−2.08

Fig. 8(b), the mean error before and after optimization is decreased from 64.59 mm to 24.61 mm, showing good stability of the model. Therefore, using SSA to optimize model parameters can significantly improve the accuracy of excavator 3D pose estimation.

The delay problem of vehicle terminals can be divided into data sampling delay, data transmission delay, and data calculation delay. (1) Due to vehicle terminals transmitting data through wired channels, theoretically, the data transmission delay on the Pad can be controlled at nanosecond levels. (2) The parameter optimization process of excavator 3D pose estimation is a work that has been debugged before the excavation task begins. In practical applications, the optimized parameters are directly used for coordinate conversion. We calculated and processed the collected 80 sets of data on the Pad, taking a total of 28 milliseconds, with an average calculation time of 0.35 milliseconds for each set of data. Therefore, the delay caused by the algorithm calculation can also be ignored. (3) The sampling frequency of the sensor in this paper is 20 Hz, so the sampling delay is 0.05 s. Therefore, the total delay of the system response on the vehicle terminal is mainly determined by the sampling frequency of the system, which is approximately 0.05 s. In comparison, the system response delay at the remote monitoring client is mainly caused by wireless data transmission, with a delay of about 1.2 seconds during the use of the system. This time delay is completely acceptable for remote monitoring.

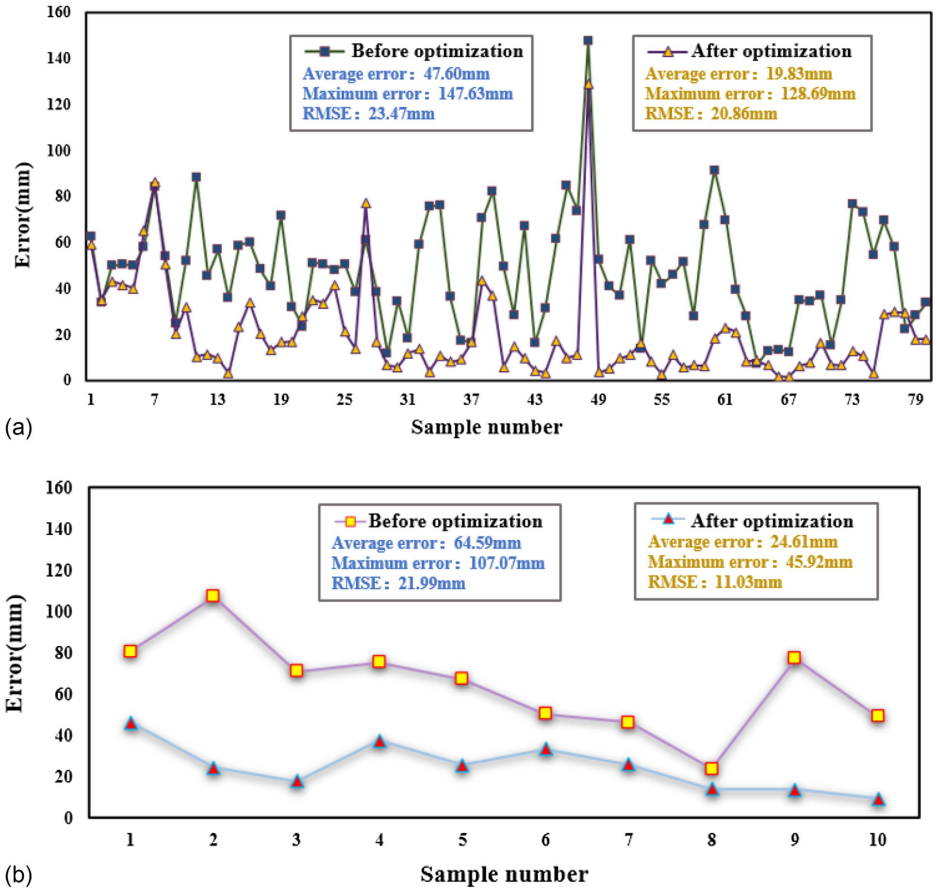


Fig. 8. Optimization results of 3D pose estimation based on SSA: (a) comparing the results before and after the parameter optimization of the training sample; and (b) comparing the results before and after the parameter optimization of the test sample.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

System Field Application

The system is applied in a sewage pipe network project in Shan Dong, China, which includes three roads to be excavated, with a total excavation length of 1,760 m. Fig. 9(a) is the project overview interface of the remote monitoring client that can remotely monitor the overall situation of the work area and help the project management personnel master the number of excavator machinery, warning times, and the location of each excavator in the work area during the current construction. Fig. 9(b) is the visual display interface of the dynamic mapping of the excavator and the BIM of underground pipes during construction, which can be displayed on both the vehicle terminal and the remote monitoring client. This interface displays the relative distance between the bucket tip and the underground pipe in real-time. Fig. 9(c) shows the bucket automatic tracking display interface at the vehicle terminal. The view is set from the top view and left view of the bucket. When the excavator enters the risk area, the color of the underground pipes model will change. The yellow indicates that the excavator has entered the low-risk area, and the red indicates that the excavator has entered the high-risk area, and the Pad will give a voice warning. At the same time, the server will send a message to the field managers.

After the excavation task is completed, the system administrator can replay the excavation process and inquire about the bucket tip trajectory and working position trajectory at the remote monitoring client. As shown in Fig. 10(a), the system can replay the excavation process in a period before and after the alarm and can assist the field management person to analyze the warning reason. Fig. 10(b) shows a graphical report of the bucket tip trajectory of the excavator for a digging operation. In this report, for every 10 cm movement of the bucket tip coordinate, the system draws a 3D space point with different colors according to the area of the bucket tip. The bucket tip trajectory is displayed in green in the safe area, yellow in the low-risk area, and red in the high-risk area to visually reflect the field excavation. During excavation, the excavator bucket is mostly in a safe area. When the bucket enters the risk area, the system will give a warning in advance, effectively avoiding the collision between the bucket and the underground pipes. Fig. 10(c) is a graphical report of the working position track of an excavator within 40 minutes, which can plot the excavator working track and provide data support for subsequent mechanical efficiency analysis.

The project needs to replace the old underground pipes, the design elevation of the trench is 50cm above the original underground pipes. The system counts the minimum relative distance between the bucket tip and the underground pipes every 1 m, and the results of K2 + 150-K2 + 200 and K3 + 750-K3 + 800 sections are shown in Fig. 11. It can be seen that during the construction, the average elevation of the trench is 50.44 cm away from the underground pipe, indicating that the construction quality meets the design requirements. Through model visual guidance and voice warning, the minimum distance between the bucket tip and the underground pipe is 23.67cm, ensuring the safety and reliability of field construction.

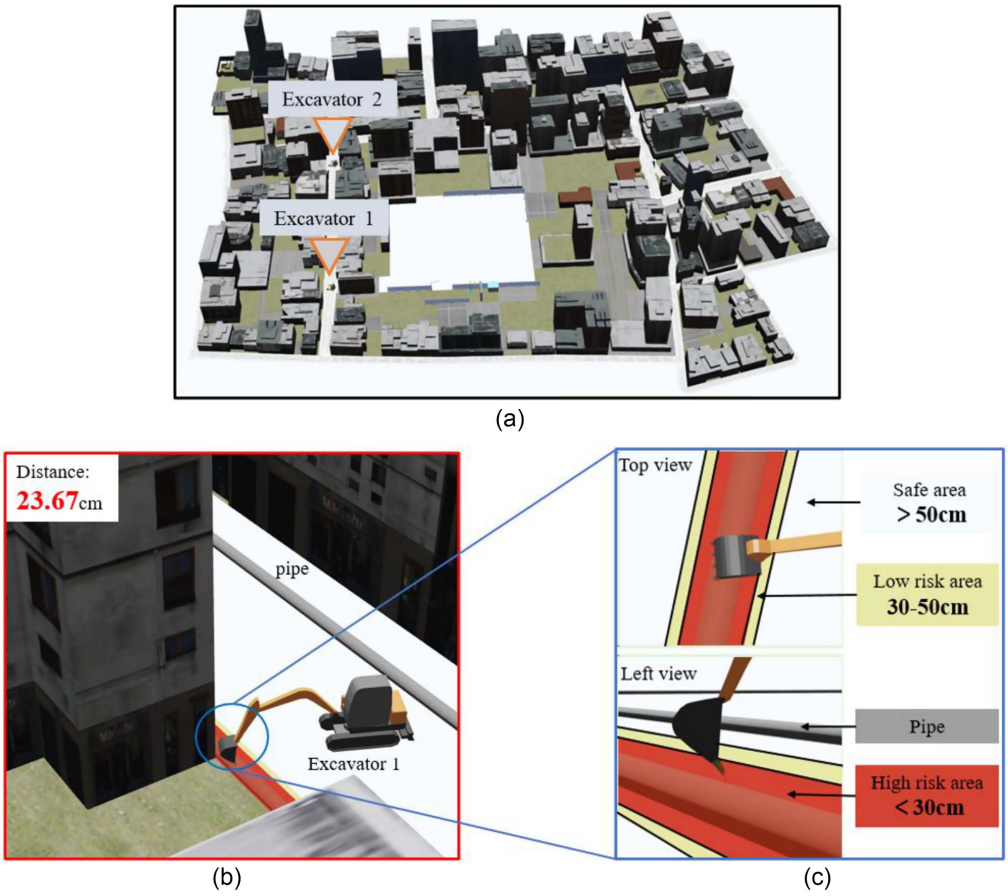


Fig. 9. System field application interface: (a) remote monitoring interface; (b) model visualization display interface; and (c) bucket automatic tracking interface.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

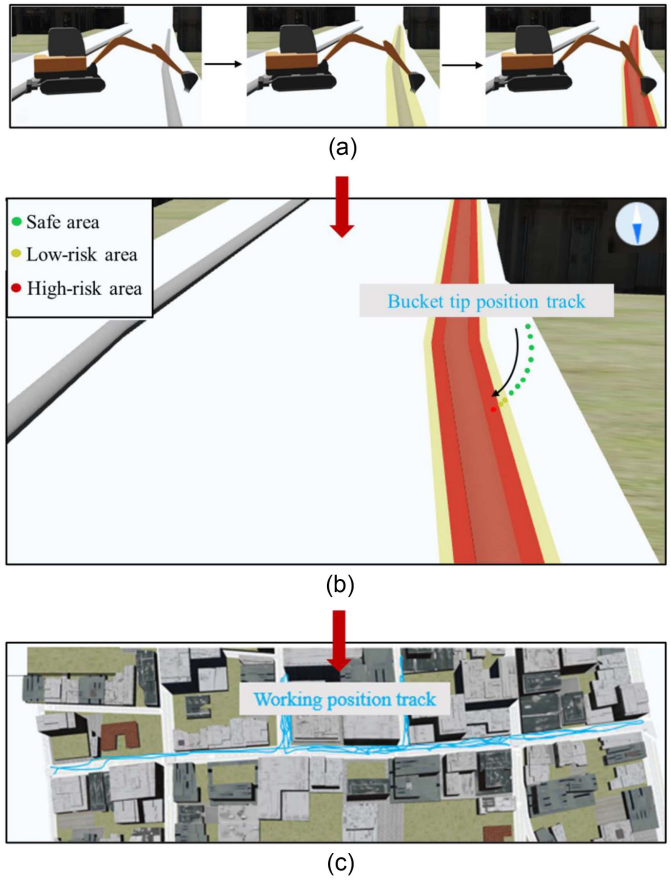


Fig. 10. Dynamic playback of the excavation process and 3D drawing of the track.

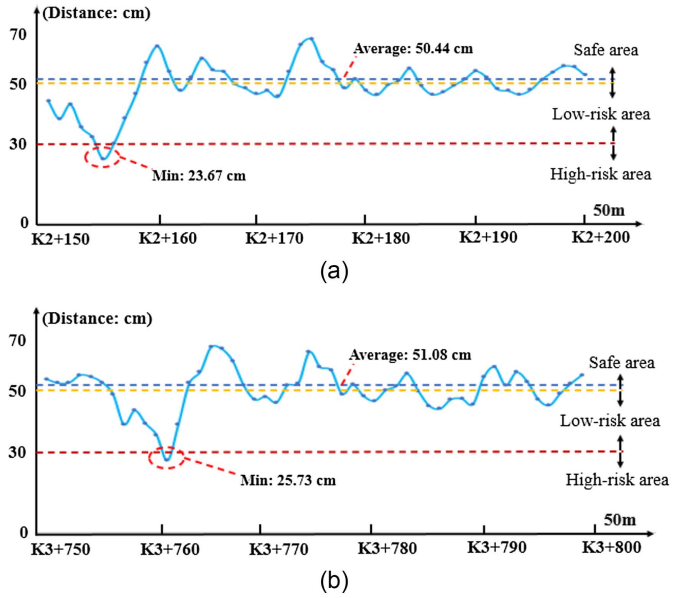


Fig. 11. Monitoring results of the minimum relative distance between bucket tip and underground pipes: (a) mileages of K2 + 150-K2 + 200; and (b) mileages of K3 + 750-K3 + 800.

Discussion

The digital twinning of urban excavation is a meaningful yet challenging task. It is meaningful in the sense that can achieve high-precision 3D pose estimation of excavators, real-time calculation of the relative distance between the bucket tip and the underground pipes, and multiterminal visualization of the calculation results to relevant personnel to improve UES. The challenge, nevertheless, is the difficulty in the accurate calibration of equipment parameters, resulting in low accuracy of 3D pose estimation, which is fatal for UES. Another challenge is how to build a system with high accuracy, fast response, and satisfactory 3D visualization effects. To tackle the challenges, this study proposes a digital twinning method of multisensory and BIM-integrated to improve UES. The contributions are three-fold.

First, a high-precision 3D pose estimation method for the excavator is proposed. Compared with the visual pose estimation method, since the excavator pose data is acquired based on IMU, there is no need to set up a camera or installation marking on the site, and it is not limited by the influence of light, distance, and object obscuration. Compared with the previous nonvisual attitude estimation methods, the excavator kinematics model proposed in this paper based on Euler transformation can obtain a coordinate transformation matrix directly from IMU, which is easy to calculate. More importantly, the SSA is innovatively applied to the parameter calibration of 3D pose estimation. The calculation accuracy of related works is mostly about 5–10 cm, and the accuracy before optimization in the paper is about 6 cm, which is similar. However, after optimization, the accuracy is raised from 64.59 mm to 24.61 mm, demonstrating its effectiveness.

Second, the paper innovatively proposes a DT design framework for urban excavation, which integrates IMU, BIM, and 3D pose estimation algorithms to realize virtual mapping of urban excavation. Previous scholars mostly carried out research on excavator 3D pose estimation and intelligent excavator guidance systems, respectively. In the aspect of intelligent excavator guidance system research, more attention was paid to the uncertain modeling of underground pipes. Relevant research lacks a complete set of closure realization methods from hardware installation, data acquisition, model establishment, pose estimation, 3D visual display, and alarm feedback.

Third, a software and hardware system has been developed, and through on-site testing, the system has achieved satisfactory application results. It can achieve 3D visual display, multiterminal real-time monitoring, dangerous behavior early warning, task query and statistics, and other functions. In particular, the delay in data transmission and computation in the system can reach the millisecond level, which can be negligible. The real-time performance of the system can reach 0.05 s, which is determined by the sampling frequency of the hardware device.

Conclusion and Prospect

In order to solve the problem of excavators colliding with underground pipes in urban excavation, this research proposes an intelligent guidance and safety early warning technology based on DT, develops an intelligent guidance and safety warning system for urban excavation, and verifies the practicability and performance of this technology through two case studies. In light of the results of this research, it can be concluded that (1) The DT can collect and aggregate real-time pose information of physical excavators with the capability to precisely estimate its 3D pose based on the kinematic model and SSA. The first case study shows that this method can effectively and stably estimate the 3D position and pose of the excavator without being constrained by the construction

Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.
Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.

environment, and the average error between the 3D pose estimated coordinates and the true 3D coordinates of the bucket tip is 24.61 mm; (2) The DT can integrate a BIM of buried utilities, urban terrains, and a 3D model of the excavators to form a dynamic virtual model that mirrors actual excavation process; (3) Based on the field excavation test of the developed intelligent guidance and safety early warning system for urban excavation, the system can realize 3D visual guidance and calculate the relative distance between the bucket tip and the underground pipes in real-time to assist the excavator operator in safe construction.

The system still has some limitations that deserve further study. The system sets the safety warning distance based on experience. In the subsequent research, the system should determine a reasonable safety warning distance by taking into account the uncertainty of the BIM model of the underground pipes through field tests. It is worth mentioning that since the coordinate system of the mechanical components of the excavator is established independently, this 3D pose estimation method can estimate the 3D pose of any point on the mechanical components, and it can be further applied to other heavy machinery in the future.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request. It mainly includes the following data: (1) field data for pose estimation; and (2) code for algorithm optimization and pose estimation.

Acknowledgments

This research was supported by the National Natural Science Foundation of China (No. 52279136).

References

Adhvaryyu, S., and P. K. Adhvaryyu. 2020. "Application of bio-inspired social spider algorithm in multi-area economic emission dispatch of solar, wind and CHP-based power system." *Soft Comput.* 24 (13): 9611–9624. https://doi.org/10.1007/s00500-019-04468-4.

Akanmu, A. A., C. J. Anumba, and O. O. Ogunseiju. 2021. "Towards next generation cyber-physical systems and digital twins for construction." *J. Inf. Technol. Constr.* 26 (Jul): 505–525. https://doi.org/10.36680/j.itcon.2021.027.

Assadzadeh, A., M. Arashpour, I. Brilakis, T. Ngo, and E. Konstantinou. 2022. "Vision-based excavator pose estimation using synthetically generated datasets with domain randomization." *Autom. Constr.* 134 (Feb): 104089. https://doi.org/10.1016/j.autcon.2021.104089.

Behzadan, A. H., S. Dong, and V. R. Kamat. 2015. "Augmented reality visualization: A review of civil infrastructure system applications." *Adv. Eng. Inf.* 29 (2): 252–267. https://doi.org/10.1016/j.aei.2015.03.005.

Croatti, A., M. Gabellini, S. Montagna, and A. Ricci. 2020. "On the integration of agents and digital twins in healthcare." *J. Med. Syst.* 44 (9): 1–8. https://doi.org/10.1007/s10916-020-01623-5.

El-bages, M. S., and W. T. Elsayed. 2017. "Social spider algorithm for solving the transmission expansion planning problem." *Electr. Power Syst. Res.* 143 (Feb): 235–243. https://doi.org/10.1016/j.epsr.2016.09.002.

Feng, C., V. R. Kamat, and H. Cai. 2018. "Camera marker networks for articulated machine pose estimation." *Autom. Constr.* 96 (Dec): 148–160. https://doi.org/10.1016/j.autcon.2018.09.004.

Golparvar-Fard, M., A. Heydarian, and J. C. Niebles. 2013. "Vision-based action recognition of earthmoving equipment using spatio-temporal features and support vector machine classifiers." *Adv. Eng. Inf.* 27 (4): 652–663. https://doi.org/10.1016/j.aei.2013.09.001.

Gülmez, B., and S. Kulluk. 2019. "Social spider algorithm for training artificial neural networks." *Int. J. Bus. Anal.* 6 (4): 32–49. https://doi.org/10.4018/IJBAN.2019100103.

Hosamo, H. H., A. Imran, J. Cardenas-Cartagena, P. R. Svennevig, K. Svidt, and H. K. Nielsen. 2022. "A review of the digital twin technology in the AEC-FM industry." In *Advances in civil engineering*, 1–17. Cairo, Egypt: Hindawi. https://doi.org/10.1155/2022/2185170.

Huang, S., K. Yang, H. Xiao, P. Han, J. Qiu, L. Peng, D. Liu, and K. Luo. 2022. "A new head pose tracking method based on stereo visual SLAM." *J. Visual Commun. Image Represent.* 82 (Jan): 103402. https://doi.org/10.1016/j.jvcir.2021.103402.

Jiang, W., L. Ding, and C. Zhou. 2022. "Digital twin: Stability analysis for tower crane hoisting safety with a scale model." *Autom. Constr.* 138 (Jun): 104257. https://doi.org/10.1016/j.autcon.2022.104257.

Kaewunruen, S., and Q. Lian. 2019. "Digital twin aided sustainability-based lifecycle management for railway turnout systems." *J. Cleaner Prod.* 228 (Aug): 1537–1551. https://doi.org/10.1016/j.jclepro.2019.04.156.

Li, S., H. Cai, and V. R. Kamat. 2015. "Uncertainty-aware geospatial system for mapping and visualizing underground utilities." *Autom. Constr.* 53 (Apr): 105–119. https://doi.org/10.1016/j.autcon.2015.03.011.

Liang, C., K. M. Lundeen, W. McGee, C. C. Menassa, S. Lee, and V. R. Kamat. 2019. "A vision-based marker-less pose estimation system for articulated construction robots." *Autom. Constr.* 104 (Aug): 80–94. https://doi.org/10.1016/j.autcon.2019.04.004.

Liu, B., R. Bao, and F. Sui. 2021. "A fatigue damage-cumulative model in peridynamics." *Chin. J. Aeronaut.* 34 (2): 329–342. https://doi.org/10.1016/j.cja.2020.09.046.

Lundeen, K. M., S. Dong, N. Fredricks, M. Akula, J. Seo, and V. R. Kamat. 2016. "Optical marker-based end effector pose estimation for articulated excavators." *Autom. Constr.* 65 (May): 51–64. https://doi.org/10.1016/j.autcon.2016.02.003.

Luo, H., M. Wang, P. K. Wong, and J. C. P. Cheng. 2020. "Full body pose estimation of construction equipment using computer vision and deep learning techniques." *Autom. Constr.* 110 (Feb): 103016. https://doi.org/10.1016/j.autcon.2019.103016.

Madubuike, O. C., C. J. Anumba, and R. Khallaf. 2022. "A review of digital twin applications in construction." *J. Inf. Technol. Constr.* 27 (8): 145–172. https://doi.org/10.36680/j.itcon.2022.008.

Mahmood, B., S. Han, and J. Seo. 2022. "Implementation experiments on convolutional neural network training using synthetic images for 3D pose estimation of an excavator on real images." *Autom. Constr.* 133 (Jan): 103996. https://doi.org/10.1016/j.autcon.2021.103996.

Moon, S., and J. Seo. 2017. "Virtual graphic representation of construction equipment for developing a 3D earthwork BIM." *J. Civ. Eng. Manage.* 23 (8): 977–984. https://doi.org/10.3846/13923730.2017.1348981.

Mostafa, K., and T. Hegazy. 2021. "Review of image-based analysis and applications in construction." *Autom. Constr.* 122 (Feb): 103516. https://doi.org/10.1016/j.autcon.2020.103516.

Panchal, K., and H. Shah. 2013. "3d face recognition based on pose correction using Euler angle method." In *Proc., Int. Conf. on Machine Intelligence and Research Advancement*, 467–471. New York: IEEE.

Perumal, L. 2014. "Euler angles: Conversion of arbitrary rotation sequences to specific rotation sequence." *Comput. Anim. Virtual Worlds* 25 (5–6): 521–529. https://doi.org/10.1002/cav.1529.

Shao, G., and M. Helu. 2020. "Framework for a digital twin in manufacturing: Scope and requirements." *Manuf. Lett.* 24 (Apr): 105–107. https://doi.org/10.1016/j.mfglet.2020.04.004.

Sharafat, A., M. S. Khan, K. Latif, W. A. Tanoli, W. Park, and J. Seo. 2021. "BIM-GIS-based integrated framework for underground utility management system for earthwork operations." *Appl. Sci.* 11 (12): 5721. https://doi.org/10.3390/app11125721.

Soltani, M. M., Z. Zhu, and A. Hammad. 2016. "Automated annotation for visual recognition of construction resources using synthetic images." *Autom. Constr.* 62 (Feb): 14–23. https://doi.org/10.1016/j.autcon.2015.10.002.

Soltani, M. M., Z. Zhu, and A. Hammad. 2017. "Skeleton estimation of excavator by detecting its parts." *Autom. Constr.* 82 (Oct): 1–15. https://doi.org/10.1016/j.autcon.2017.06.023.

Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Su, X., S. Talmaki, H. Cai, and V. R. Kamat. 2013. "Uncertainty-aware visualization and proximity monitoring in urban excavation: A geospatial augmented reality approach." *Visualization Eng.* 1 (1): 1–13. https://doi.org/10.1186/2213-7459-1-2.

Sun, D., C. Ji, S. Jang, S. Lee, J. No, C. Han, J. Han, and M. Kang. 2020. "Analysis of the position recognition of the bucket tip according to the motion measurement method of excavator boom, stick and bucket." *Sensors* 20 (10): 2881. https://doi.org/10.3390/s20102881.

Sun, J., and S. Lu. 2020. "An improved single shot multibox for video-rate head pose prediction." *IEEE Sens. J.* 20 (20): 12326–12333. https://doi.org/10.1109/JSEN.2020.2999625.

Sung, H., N. Jung, S. Huang, and J. Kim. 2019. "Application of social spider algorithm to optimize train energy." *J. Electr. Eng. Technol.* 14 (1): 519–526. https://doi.org/10.1007/s42835-018-00016-6.

Talmaki, S., and V. R. Kamat. 2012. "Real-time hybrid virtuality for prevention of excavation related utility strikes." *J. Comput. Civ. Eng.* 28 (3): 04014001. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000269.

Talmaki, S., V. R. Kamat, and H. Cai. 2013. "Geometric modeling of geospatial data for visualization-assisted excavation." *Adv. Eng. Inf.* 27 (2): 283–298. https://doi.org/10.1016/j.aei.2013.01.004.

Tang, J., H. Luo, W. Chen, P. K. Wong, and J. C. P. Cheng. 2022. "IMU-based full-body pose estimation for construction machines using kinematics modeling." *Autom. Constr.* 138 (Jun): 104217. https://doi.org/10.1016/j.autcon.2022.104217.

Tanoli, W. A., J. W. Seo, A. Sharafat, and S. S. Lee. 2018. "3D design modeling application in machine guidance system for earthwork operations." *KSCE J. Civ. Eng.* 22 (12): 4779–4790. https://doi.org/10.1007/s12205-018-0309-y.

Tanoli, W. A., A. Sharafat, J. Park, and J. W. Seo. 2019. "Damage prevention for underground utilities using machine guidance." *Autom. Constr.* 107 (Nov): 102893. https://doi.org/10.1016/j.autcon.2019.102893.

Vahdatikhaki, F., A. Hammad, and H. Siddiqui. 2015. "Optimization-based excavator pose estimation using real-time location systems." *Autom. Constr.* 56 (Aug): 76–92. https://doi.org/10.1016/j.autcon.2015.03.006.

Wu, S., L. Hou, G. K. Zhang, and H. Chen. 2022. "Real-time mixed reality-based visual warning for construction workforce safety." *Autom. Constr.* 139 (Jul): 104252. https://doi.org/10.1016/j.autcon.2022.104252.

Xu, J., and H. Yoon. 2019. "Vision-based estimation of excavator manipulator pose for automated grading control." *Autom. Constr.* 98 (Feb): 122–131. https://doi.org/10.1016/j.autcon.2018.11.022.

Yu, J. J. Q., and V. O. K. Li. 2015. "A social spider algorithm for global optimization." *Appl. Soft Comput.* 30 (Jan): 614–627. https://doi.org/10.1016/j.asoc.2015.02.014.

Zeng, Y., and H. Guo. 2017. "An Euler angle calculation method for Tailsitter UAV." In *Proc., Chinese Automation Congress (CAC)*, 814–819. New York: IEEE.

Zhao, J., Y. Hu, and M. Tian. 2021. "Pose estimation of excavator manipulator based on monocular vision marker system." *Sensors* 21 (13): 4478. https://doi.org/10.3390/s21134478.

Zou, J., and H. Kim. 2007. "Using hue, saturation, and value color space for hydraulic excavator idle time analysis." *J. Comput. Civ. Eng.* 21 (4): 238–246. https://doi.org/10.1061/(ASCE)0887-3801(2007)21:4(238).

Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.
Downloaded from ascelibrary.org by City University of Hong Kong on 09/15/26.